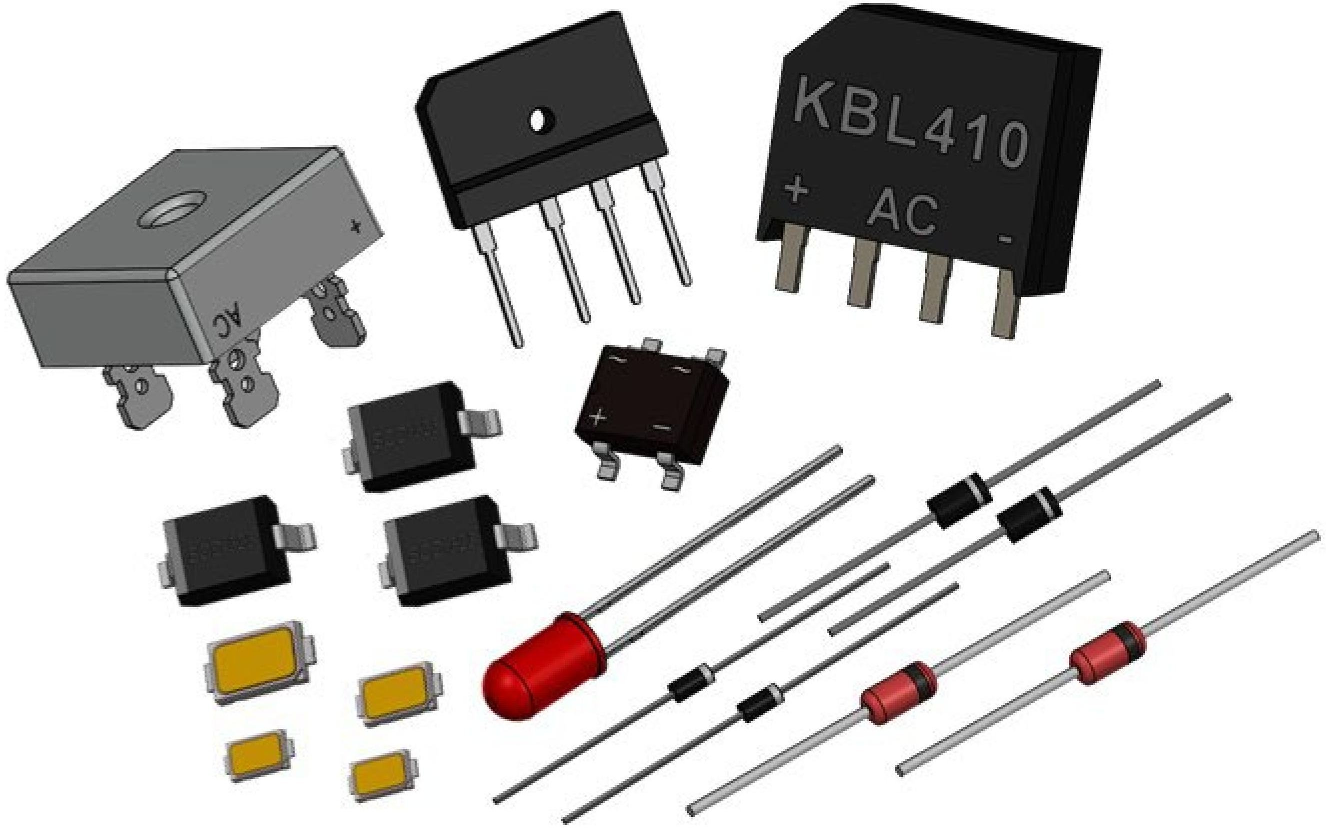
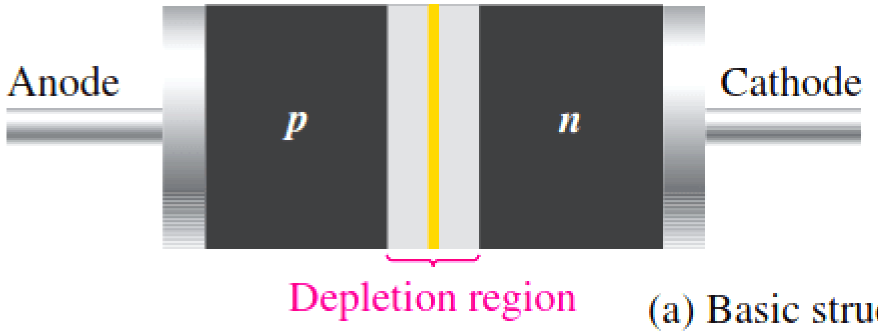
# Diodes

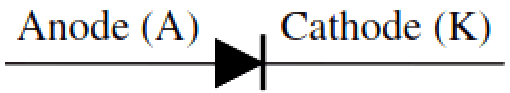


- Structure of a diode
  - Diode consist of single pn junction. The terminal connected to the p-type semiconductor is labeled as Anode and the terminal connected to the n-type semiconductor is named as Cathode.

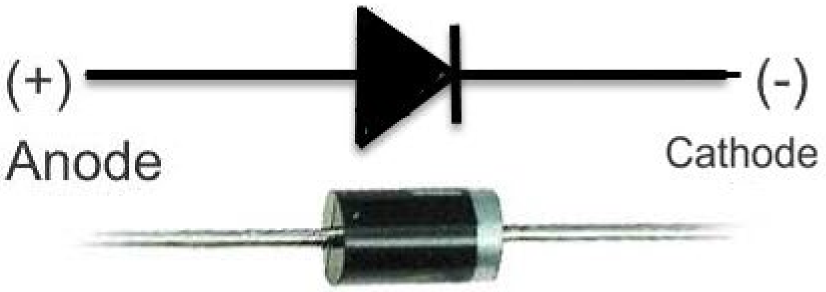
**Diode**



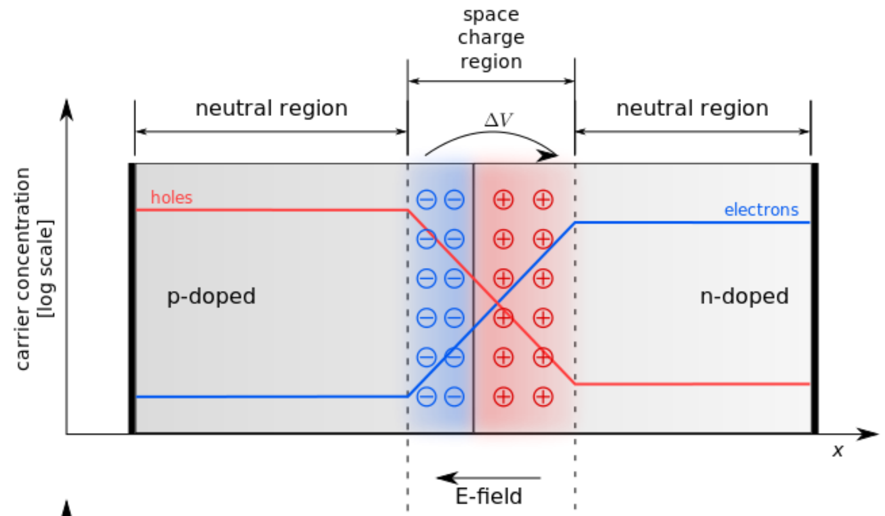

(a) Basic structure



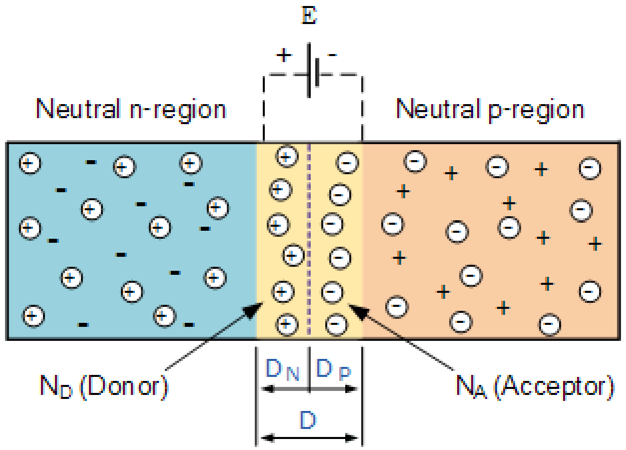

(b) Symbol



- PN junction
  - When the N-type semiconductor and P-type semiconductor materials are first joined together a very large density gradient exists between both sides of the PN junction. The result is that some of the free electrons from the donor impurity atoms begin to migrate across this newly formed junction to fill up the holes in the P-type material producing negative ions.
  - However, because the electrons have moved across the PN junction from the N-type silicon to the P-type silicon, they leave behind positively charged donor ions on the negative side and now the holes from the acceptor impurity migrate across the junction in the opposite direction into the region where there are large numbers of free electrons.
  - This charge transfer of electrons and holes across the PN junction is known as diffusion. The width of these P and N layers depends on how heavily each side is doped with acceptor density $N_A$, and donor density $N_D$, respectively.



- This process continues back and forth until the number of electrons which have crossed the junction have a large enough electrical charge to repel or prevent any more charge carriers from crossing over the junction. Eventually a state of equilibrium (electrically neutral situation) will occur producing a “potential barrier” zone.
- Since no free charge carriers can rest in a position where there is a potential barrier, the regions on either sides of the junction now become completely depleted of any more free carriers in comparison to the N and P type materials further away from the junction. This area around the PN Junction is now called the Depletion Layer.
- The total charge on each side of a PN Junction must be equal and opposite to maintain a neutral charge condition around the junction. If the depletion layer region has a distance D, it therefore must therefore penetrate into the silicon by a distance of $D_p$ for the positive side, and a distance of $D_n$ for the negative side giving a relationship between the two of: $D_p*N_A = D_n*N_D$ in order to maintain charge neutrality also called equilibrium.



- Potential barrier (or junction potential) depends on the semiconductor type. For a Si it is typically 0.6 - 0.7 V and for Ge it is 0.2 - 0.3 V.

- Calculating potential barrier
  - Potential barrier $V_B$ is given by,

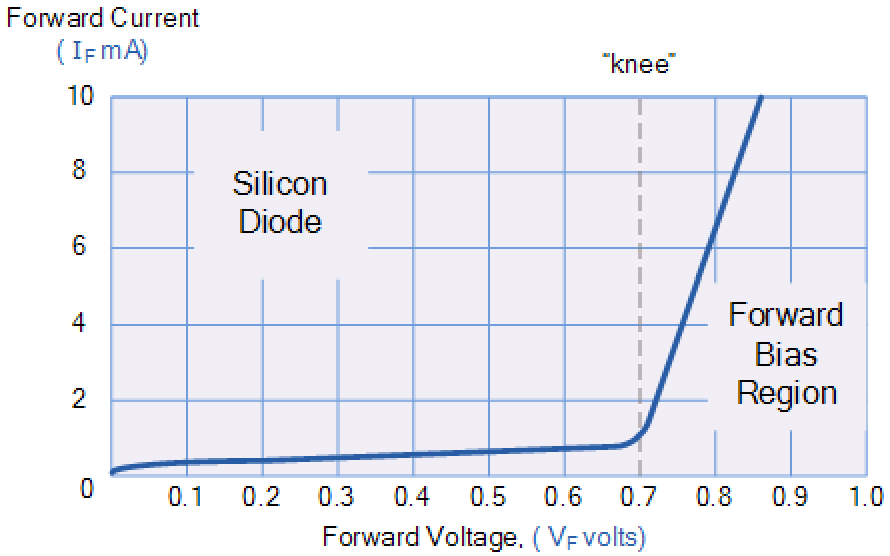
$$V_B = \left(\frac{kT}{q}\right) \ln\left(\frac{N_A N_D}{n_i^2}\right)$$

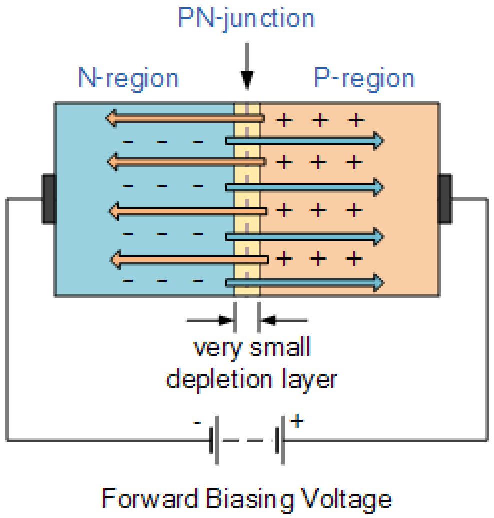
  - Where, k is the Boltzmann constant ($1.3807 \times 10^{-23}$ $JK^{-1}$) T is the temperature in kelvin, $N_A$ and $N_D$ are the doping concentration in acceptor and donor respectively, $n_i$ is the carrier density of intrinsic semiconductor. At room temperature (300K) the value of $kT/q$ is 0.026 V.

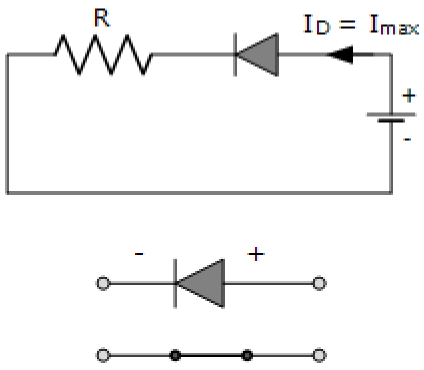
- I-V characteristic of a diode
  - Diode can be connected to a voltage source in two different configurations. If the anode has a higher potential than the cathode, it is called forward biased. If the cathode has a higher voltage than the anode, it is called the reverse biased.

- Forward Biased PN Junction Diode
  - When a diode is connected in a Forward Bias condition, a negative voltage is applied to the N-type material and a positive voltage is applied to the P-type material. If this external voltage becomes greater than the value of the potential barrier, approx. 0.7 volts for silicon and 0.3 volts for germanium, the potential barriers opposition will be overcome and current will start to flow.
  - This is because the negative voltage pushes or repels electrons towards the junction giving them the energy to cross over and combine with the holes being pushed in the opposite direction towards the junction by the positive voltage. This results in a characteristics curve of zero current flowing up to this voltage point, called the “knee” on the static curves and then a high current flow through the diode with little increase in the external voltage.

- The application of a forward biasing voltage on the junction diode results in the depletion layer becoming very thin and narrow which represents a low impedance path through the junction thereby allowing high currents to flow.

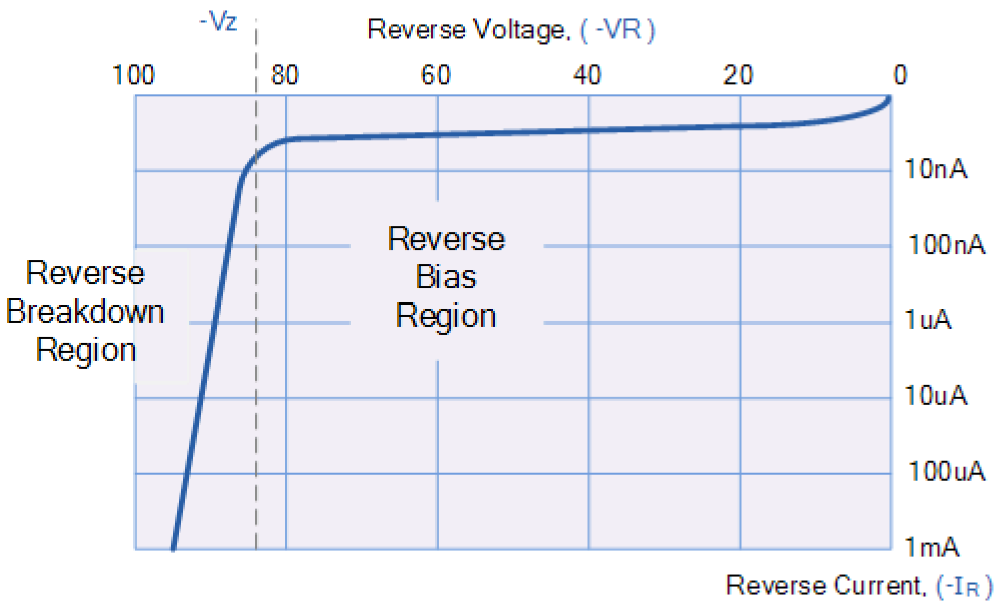


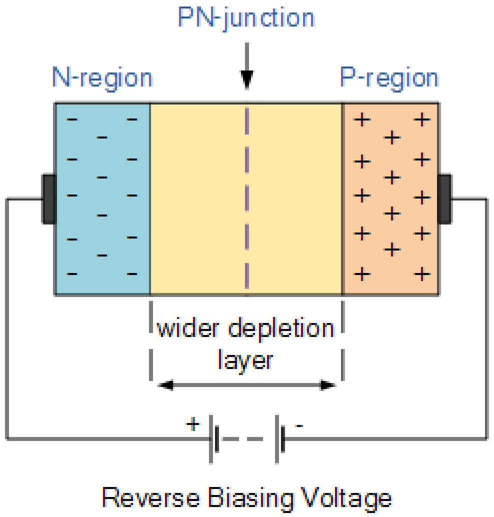



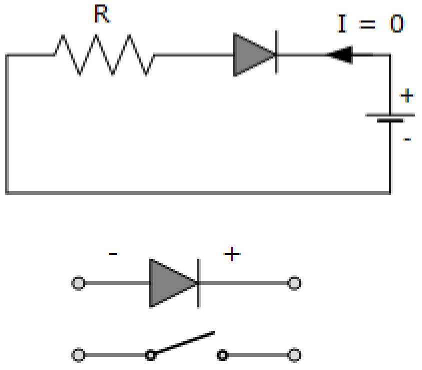




- Since the diode can conduct "infinite" current above this knee point as it effectively becomes a short circuit, therefore resistors are used in series with the diode to limit its current flow. Exceeding its maximum forward current specification causes the device to dissipate more power in the form of heat than it was designed for resulting in a very quick failure of the device.

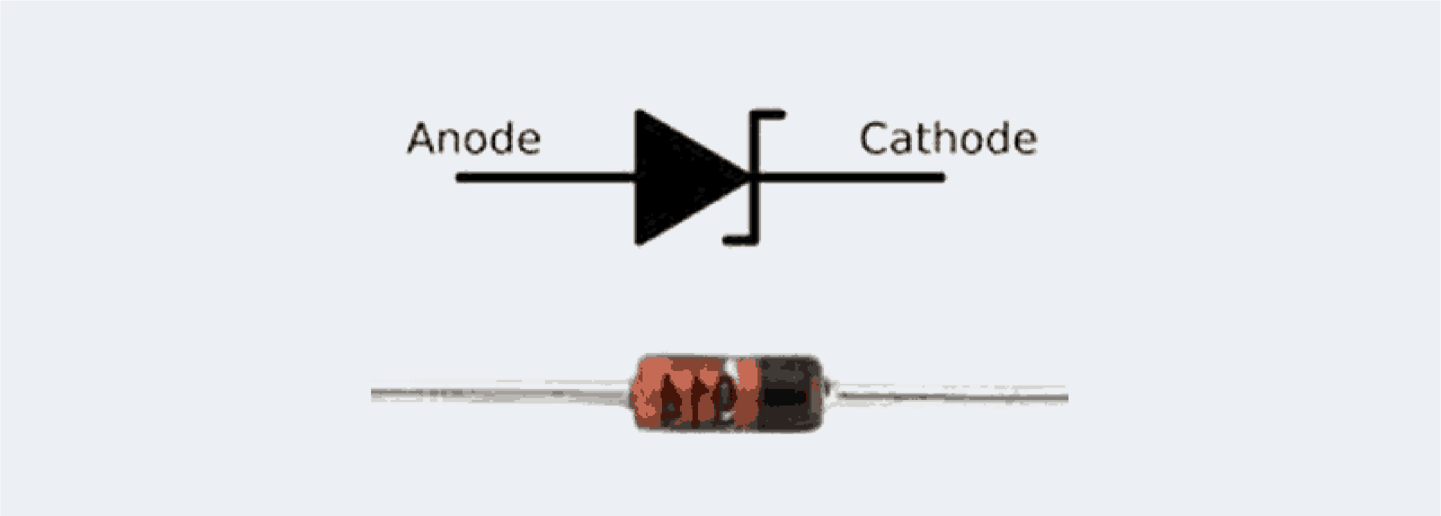
- Reverse Biased PN Junction Diode
  - When a diode is connected in a Reverse Bias condition, a positive voltage is applied to the N-type material and a negative voltage is applied to the P-type material. The positive voltage applied to the N-type material attracts electrons towards the positive electrode and away from the junction, while the holes in the P-type end are also attracted away from the junction towards the negative electrode.
  - The net result is that the depletion layer grows wider due to a lack of electrons and holes and presents a high impedance path, almost an insulator and a high potential barrier is created across the junction thus preventing current from flowing through the semiconductor material.



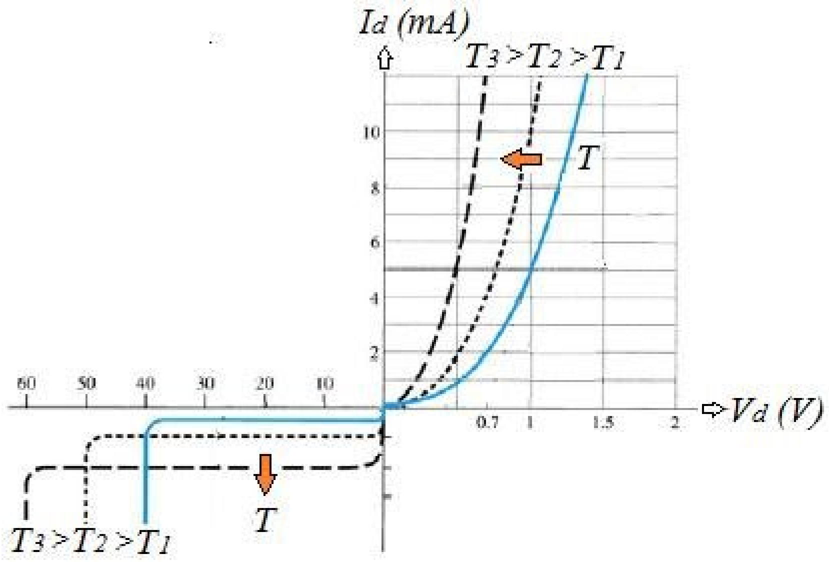



- This condition represents a high resistance value to the PN junction and practically zero current flows through the junction diode with an increase in bias voltage. However, a very small reverse leakage current does flow through the junction which can normally be measured in micro-amperes, ( μA ).
- One final point, if the reverse bias voltage $V_r$ applied to the diode is increased to a sufficiently high enough value, it will cause the diode's PN junction to overheat and fail. This may cause the diode to become shorted and will result in the flow of maximum circuit current, and this shown as a step downward slope in the reverse static characteristics curve.
- Sometimes this failing effect has practical applications in voltage stabilizing circuits where a series limiting resistor is used with the diode to limit this reverse breakdown current to a preset maximum value thereby producing a fixed voltage output across the diode. These types of diodes are commonly known as Zener Diodes



- Effect of temperature on IV characteristics
  - Current-Voltage (I-V) characteristic of a junction diode is largely depend on the temperature. As temperature increases, additional electron-hole pair generation take place and both forward and reverse bias characteristics curves will be altered.




  - When temperature increases, the reverse saturation current ($I_s$) also increases due to the increase in minority carriers.
  - Forward current increases rapidly at lower forward voltage. When temperature of the junction diode is increased the knee voltage reduces and forward current starts increasing.